Rapid Revision Summary

Last-minute review · condensed from the Paper 1C A Guide (17 papers, Jan 2019 – Jun 2026)*

THE PAPER

2 hours · 110 marks · 9–12 questions · ≈ 1 mark per minute. Easy recall first; final questions are multi-step calculations or a 6-mark practical. Every paper: moles and organic (17/17); bonding (16); rates, tests, states of matter (14). MCQ distractors repeat.

TOP 12 MARK-LOSERS

- Covalent bond: shared pair and the **nuclei** (not "nucleus").
- Simple molecular: **weak intermolecular forces**. Breaking covalent bonds = 0.
- Giant structures: never mention intermolecular forces or molecules.
- "More energy" is ignored: "**a large amount of energy** to break the strong bonds".
- Rates: "**more collisions per unit time**". Energy only for temperature.
- Polystyrene **reduces** heat loss ("prevents" rejected).
- ΔH needs a sign. Thermometer to 1 d.p. (16.0). Never 1 s.f.
- Zinc **oxidises/corrodes instead of iron** ("zinc rusts" rejected). Rust = hydrated iron(III) oxide.
- -ine = element, -ide = ion.
- Isomers: same **molecular** formula. Homologous: **trend** in physical properties.
- Bromine water: orange → colourless ("clear" ignored, "discoloured" rejected).
- HNO_3 before $AgNO_3$; HCl before $BaCl_2$. Heat to crystallisation point, not dryness.

STATES, DIFFUSION, PURITY

Change	Name
solid → liquid / liquid → solid	melting / freezing
liquid → gas / gas → liquid	evaporation (boiling) / condensation
solid → gas	sublimation (I_2, CO_2)

State	Arrangement	Movement
solid	regular, close	vibrate in fixed positions
liquid	close, random	move around each other
gas	random, far apart	fast, random

EVAPORATES FASTER WHEN HEATED (2)
Particles gain kinetic energy, so more have enough energy to overcome intermolecular forces and escape from the surface.

- NH_3/HCl tube: white ring nearer HCl; HCl higher M_r (36.5 vs 17), moves slower. Slow because particles collide with air particles.
- Pure: sharp, fixed mp. Impure: lower, over a range. Pure water boils at 100 °C (physical test). Anhydrous $CuSO_4$ white → blue shows water present, not purity.

SEPARATION AND CHROMATOGRAPHY

Separate	Method
insoluble solid + liquid	filtration (residue)
soluble solid from solution	crystallisation
solvent from solution	simple distillation
liquids, different bp	fractional distillation
dyes	paper chromatography

CHROMATOGRAPHY METHOD (5)
Pencil start line near bottom → small spot of each ink → solvent in beaker **below** the line → stand paper in → leave until solvent near top, mark solvent front, dry.

- Ink line dissolves and moves up; solvent above spots dissolves dyes into the beaker.
- R_f = spot distance ÷ solvent distance (from start line), always < 1. Same R_f = same dye. A spot that has not moved may still hold several dyes.
- Reverse: 0.72 × 120 mm = 86 mm.

ATOMS AND ISOTOPES

ISOTOPES (2)
Same number of protons, different neutrons. Same chemical properties: same electronic configuration.

- Proton 1, +1; neutron 1, 0; electron 1/1836, −1. Group = outer electrons; period = shells.
- A_r = Σ(mass × %) ÷ 100: Br 79.9, Mg 24.3, B 10.8. Unrounded = 2 of 3.
- 1 mol = 6.0×10^{23}. Electrons in 1 mol Mg = $12 \times 6.0 \times 10^{23} = 7.2 \times 10^{24}$. Atoms in 1 mol C_{60} = 3.6×10^{25}.
- Metal oxides basic; non-metal oxides acidic.

GROUPS 1, 7, 0

- Group 1 + water: fizzing, floats, moves, gets smaller; Na melts into ball; K lilac flame. $2Li + 2H_2O \rightarrow 2LiOH + H_2$. OH^-: UI blue/purple, phenolphthalein pink. Reactivity increases down.
- Equal masses: 0.25 g K is fewer moles than 0.25 g Na → less H_2.
- F pale yellow gas · Cl pale green gas · Br red-brown liquid (orange aq) · I grey-black solid (brown aq) · At black solid. Reactivity decreases down; bp increases (stronger intermolecular forces).
- Group 0: full outer shell, unreactive. Helium in airships: low density, not flammable.

DISPLACEMENT (6)

Br_2 + KCl: stays orange → Cl more reactive than Br. Br_2 + KI: turns brown, $Br_2 + 2KI \rightarrow 2KBr + I_2$ → Br more reactive than I. Order Cl > Br > I.

Redox: chlorine gains electrons (reduced); **iodide ions** lose electrons (oxidised).

AIR AND POLLUTION

- N_2 78%, O_2 21%, Ar 1%, CO_2 0.04%. Air is a mixture: no formula.
- % O_2 = used ÷ (syringe + flask) × 100: (90 − 22) ÷ 350 = 19.4%.
- Hot copper turns black; constant reading = all O_2 used. Low result: not left long enough, leak, not cooled.
- Iron method: iron wet (water needed); powder = larger surface area.

Gas	Problem
CO_2	greenhouse gas → global warming
CO (incomplete combustion)	toxic: reduces blood's oxygen capacity
SO_2 (S in fuel)	acid rain: kills fish/plants, corrodes limestone
NO_x (N_2 + O_2 from air, high temp)	acid rain

Nitrogen comes from the air, not petrol. No ozone-layer links.

FORMULAE AND MOLES

Ions: NH_4^+, Ag^+, Fe^{3+}, Al^{3+}, NO_3^-, SO_4^{2-}, CO_3^{2-}, PO_4^{3-}, N^{3-}. e.g. $(NH_4)_2SO_4$, $Ca_3(PO_4)_2$ (13 atoms), Mg_3N_2, $Al_2(CO_3)_3$.

Type	Example
% by mass	Ca in $Ca_3(PO_4)_2$: 120 ÷ 310 = 38.7%
empirical	Pb 90.7, O 9.3 → 0.438 : 0.581 = 3 : 4 → Pb_3O_4 (÷ A_r, never atomic no.)
molecular	C_2H_4Cl (63.5), M_r 127 → $C_4H_8Cl_2$
reacting mass	28 g Fe = 0.5 mol → 0.25 mol Fe_2O_3 → 40 g
tonnes	892 t PbO ÷ 223 = 4 Mmol → 2 Mmol CO_2 → 88 t
limiting	$2Mg + O_2$: 0.50 Mg, 0.20 O_2 → O_2 limits → 0.40 mol MgO
excess left	0.025 mol HNO_3 uses 0.0125 mol Mg = 0.30 g; 0.75 − 0.30 = 0.45 g
% yield	21 ÷ 28 = 75%; > 100% = product wet
combustion	396 g CO_2 = 9, 180 g H_2O = 10 → C_9H_{20}; O_2 = 14
find A_r	0.32 g O = 0.020 mol; 4.14 ÷ 0.020 = 207 → Pb

BONDING AND STRUCTURE

IONIC

Bond: strong electrostatic attraction between oppositely charged ions. Mg 2.8.2 → 2.8; Cl 2.8.7 → 2.8.8. Brackets + charges. Sharing = 0.

High mp (3): giant lattice, strong electrostatic forces between oppositely charged ions, large amount of energy to overcome. MgO > NaCl: higher charges.

Conducts molten/aqueous (ions move), not solid (ions fixed).

Structure	mp	Conducts	Examples
simple molecular	low	no	CH_4, CO_2, halogens, C_{60}
giant covalent	very high	no (graphite yes)	diamond, graphite, SiO_2
giant ionic	high	molten/aq only	NaCl, MgO, PbS
metallic	high	yes	Cu

- Dot-and-cross: NH_3 needs the lone pair on N; N_2 triple bond + lone pair each; O_2 double bond.
- Hexane > butane bp: larger molecule, stronger intermolecular forces.
- $CaBr_2$ vs caffeine (5): strong electrostatic attractions vs weak intermolecular forces; much more energy.

DIAMOND (3) · GRAPHITE (6)

Diamond: giant covalent, each C bonded to 4, many strong covalent bonds, large energy → hard, high mp; no delocalised electrons.

Graphite: each C bonded to 3 → layers → **weak forces between layers** → slide → soft; one **delocalised electron** per C, free to move → conducts. Not "intermolecular"; no ions.

C_{60}: simple molecular, weak intermolecular forces → far lower mp.

REACTIVITY AND REDOX

K > Na > Li > Ca > Mg > Al > (C) > Zn > Fe > (H) > Cu > Ag > Au

Experiment	Observation
Mg in $CuSO_4$	brown solid; blue → colourless; warms
Fe in $CuSO_4$	pink-brown coating; blue → pale green
Zn in $FeSO_4$	dark grey coating; green → colourless
Cu in $MgSO_4$	no change

- Bigger temperature rise = more reactive. Control concentration, volume, mass, surface area, start T, stirring.
- OIL RIG. Oxidising agent gains electrons (Fe^{2+}, not "iron"); reducing agent loses them.
- Thermite $2Al + Fe_2O_3 \rightarrow 2Fe + Al_2O_3$: Al oxidised, exothermic.
- H_2 reduction: burn excess H_2; keep H_2 flowing until cool (no re-oxidation); heat to constant mass.
- Rust needs water + oxygen. Paint = barrier. Galvanising works even when scratched.

ACIDS AND SALTS

Indicator	Acid	Alkali
litmus	red	blue
phenolphthalein	colourless	pink
methyl orange	red	yellow
universal	red/orange	blue/purple

- Acid = H^+ donor; base = H^+ acceptor; alkali = soluble base giving OH^-. NH_3 pH ≈ 11.
- Insoluble: AgCl, $PbCl_2$, $BaSO_4$, $CaSO_4$, $PbSO_4$, most carbonates and hydroxides. All nitrates soluble.

PURE DRY CRYSTALS (5–6)

Warm acid, add solid until in excess (solid remains, fizzing stops) → filter → heat filtrate to crystallisation point (crystals on glass rod) → cool → filter → dry between filter papers / warm oven. Dryness = M1 only.

Excess solid: all acid reacts; easily filtered off.

Insoluble salt: mix two solutions → filter → wash → dry. $Pb(NO_3)_2 + 2KCl \rightarrow PbCl_2(s) + 2KNO_3$.

CHEMICAL TESTS

Test	Result
flame: Li^+ Na^+ K^+ Ca^{2+} Cu^{2+}	red, yellow, lilac, orange-red, blue-green
NaOH: Cu^{2+} Fe^{2+} Fe^{3+}	blue, green, red-brown ppt
NH_4^+: NaOH, warm, damp red litmus	turns blue
HNO_3 + $AgNO_3$: Cl^- Br^- I^-	white, cream, yellow ppt
HCl + $BaCl_2$: SO_4^{2-}	white ppt
acid; gas into limewater: CO_3^{2-}	fizz; milky
H_2 / O_2 / Cl_2	pop / relights / damp litmus bleached
water: anhydrous $CuSO_4$	white → blue

UNKNOWN SALT (6)

Cation test → observation → conclusion; **fresh sample**; anion test → observation → conclusion. NH_4Cl vs $(NH_4)_2SO_4$: test only the anions.

ENERGETICS

CALCULATION (4–5)

Q = mcΔT (m = water/solution, 25 + 25 = 50 g) → n = moles burned or limiting → ΔH = −Q ÷ n ÷ 1000 kJ/mol, with sign.

e.g. 100 g, ΔT 12 °C, 0.23 g ethanol: Q = 5040 J, n = 0.0050 → −1008 kJ/mol.

Reverse: ΔT = 3780 ÷ (50 × 4.2) = 18 → max T = 19 + 18 = 37 °C.

- Exothermic: heat given out, T rises, ΔH −. Endothermic: dissolving NH_4NO_3 (+24.2 kJ/mol).
- Errors: heat loss, incomplete combustion (soot), evaporation from wick. Fix: lid, insulation, draught shield, copper can, stir.
- Two-line graph: rises (heat released); after neutralisation cooler solution lowers T; intersection = end.

RATES

COLLISION THEORY

Concentration / surface area / pressure: more frequent collisions. **No energy.**

Temperature: more kinetic energy → more collisions ≥ Ea → more frequent successful collisions.

Catalyst: alternative pathway with lower Ea; not used up.

Change	Gradient	Final
hotter / powder / catalyst	steeper	same
colder / lumps	shallower	same
half limiting reagent	shallower	half

- Curve levels off when limiting reactant used up; steepest at start (highest concentration).
- Tangent for rate at a time (Δy ÷ Δx, units cm^3/s). Mean rate = total ÷ time.
- Cotton wool plug: lets CO_2 out, keeps acid spray in. No H_2SO_4 with marble ($CaSO_4$ coats).
- Thiosulfate: time for cross to vanish; rate = 1 ÷ time. MnO_2: filter, dry, same mass.

REVERSIBLE AND HYDRATES

- ⇌ = reversible (not just "equilibrium"). $CuSO_4 \cdot 5H_2O \rightleftharpoons CuSO_4 + 5H_2O$: blue → white.
- x = mol water ÷ mol anhydrous: Na_2SO_4 0.025 : 0.25 → 10; $MgSO_4$ → 7.
- Heat to constant mass. x too low: not heated long enough.

ORGANIC

FRACTIONAL DISTILLATION + CRACKING (6)

Heated, vaporises → column hot at bottom, cool at top → condense at different heights by bp → too many long chains, demand for short → silica/alumina, 600–700 °C → shorter alkanes (fuels) + alkenes (polymers).

Fractions top → bottom: refinery gas, petrol, kerosene (aircraft), diesel, fuel oil (ships), bitumen (roads). Down: longer, higher bp, more viscous, less flammable.

- Hydrocarbon: H and C atoms **only**. Saturated: only single C–C bonds.
- Alkanes C_nH_{2n+2}; alkenes C_nH_{2n} (empirical CH_2).
- Isomers: C_4H_{10} butane/methylpropane; C_4H_8 but-1-ene, but-2-ene, methylpropene. Bent chain ≠ isomer.
- Incomplete combustion: CO and/or C + H_2O.

ETHANE VS ETHENE WITH BROMINE (5)

Ethane: substitution, needs UV, stays orange; $C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$. Ethene: addition, room temp, orange → colourless; $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$.

- Polymer repeat unit: no C=C, extension bonds through brackets, n after. PVC $–CH_2–CHCl–$.
- Landfill: non-biodegradable, fills sites. Burning: CO_2 (greenhouse), toxic CO, HCl.

PRACTICAL AND FINAL CHECKS

- Plot ± half square; smooth curve or ruled line; never join dots; draw read-off lines.
- "Directly proportional" only for a straight line through the origin.
- Explain = linked with "so/because". Count marks = count points.
- Decide giant or simple first. Colours: start → end. Units, sign, s.f. on the answer line.